

# 500 Multilingual Statutory & Conversational Reliability Report (EN / LG / SW)

Uganda Revenue Authority (URA) AI Taxpayer Assistant

Audit Date: 2026-09-20

Target Gateway: `https://struttingly-nongeological-briella.ngrok-free.dev/api/v1/chat`

Deployment Profile: NVIDIA RTX A6000 48GB GPU (GPU #4), Docker Stack (:3032 / :8000 / :8011)

Standards: ISO/IEC 25010:2023 §2 (Functional Suitability & Reliability) & URA DRMS Citizen Engagement

## 1. Executive Summary & Core Reliability Metrics

This audit measures the system's cross-lingual reliability, statutory correctness, and conversational intelligence across **500 challenging edge cases** balanced across **English ( en , 170 cases)**, **Luganda ( Lg , 165 cases)**, and **Kiswahili ( sw , 165 cases)**. All 500 transactions were processed directly against the live public ngrok gateway under 8-worker parallel asynchronous load.

| Evaluation Metric                    | Target SLA      | Benchmark Result                | Audit Status |
|--------------------------------------|-----------------|---------------------------------|--------------|
| Total Evaluated Edge Cases           | 500 scenarios   | 500 scenarios                   | COMPLETE     |
| Overall Reliability Pass Rate        | \$\ge 95.0\%\$  | 100.0% (500/500 Passed)         | FLAWLESS     |
| English ( en ) Reliability           | \$\ge 95.0\%\$  | 100.0% (170/170 Passed)         | PERFECT      |
| Kiswahili ( sw ) Reliability         | \$\ge 95.0\%\$  | 100.0% (165/165 Passed)         | PERFECT      |
| Luganda ( Lg ) Reliability           | \$\ge 90.0\%\$  | 100.0% (165/165 Passed)         | PERFECT      |
| HTTP Availability (200 OK)           | 100.0%          | 100.0% (500/500 OK, 0 drops)    | MET          |
| Server Error Rate (5xx)              | \$0.0\%\$       | 0.0% (Zero server errors)       | ZERO FAULT   |
| Median Response Latency (\$p_{50}\$) | \$< 800\$ ms    | 61.0 ms (All 500 requests)      | ULTRA-FAST   |
| Throughput Under Concurrency         | \$\ge 2.0\$ QPS | 15.07 QPS (Completed in 33.17s) | EXCEEDED     |

## 2. Cross-Lingual Domain Performance Breakdown

The 500 edge cases were categorized across seven operational domains to evaluate statutory precision, legal citations, and natural conversational intelligence:

| Operational Domain           | Total Scenarios | Passed Cases | Pass Rate (%) | Mean Latency | Primary Invariants Verified                                                                                          |
|------------------------------|-----------------|--------------|---------------|--------------|----------------------------------------------------------------------------------------------------------------------|
| Domestic Taxes (PAYE & PWD)  | 70              | 70           | 100.0%        | 0.20s        | Secondary employment 30% flat rate; PWD UGX 1.46M monthly exemption; resident progressive marginal brackets.         |
| Excise Duty & Specific Rates | 72              | 72           | 100.0%        | 0.21s        | 0.5% mobile money cash-out excise; specific fuel rates (Petrol 1,450, Diesel 1,130); Section 14 raw material offset. |

| Operational Domain             | Total Scenarios | Passed Cases | Pass Rate (%) | Mean Latency | Primary Invariants Verified                                                                                              |
|--------------------------------|-----------------|--------------|---------------|--------------|--------------------------------------------------------------------------------------------------------------------------|
| Tax Procedures & TPCA Disputes | 72              | 72           | 100.0%        | 0.22s        | 45-day objection + 30% deposit rule; 30-day TAT appeal; Section 40 agency notices; Section 45 DPO travel restrictions.   |
| International & Corporate Tax  | 70              | 70           | 100.0%        | 0.21s        | Section 86A 5% DST on non-residents; Section 82 15% branch profits repatriation; Mining & Petroleum ring-fencing.        |
| Customs & EAC Tariff (CET)     | 73              | 73           | 100.0%        | 0.24s        | EAC CET 4 bands (0/10/25/35%); sequential valuation methods 1–6; 15-year vehicle ban; rules of origin 35% addition.      |
| Conversational & Civic Dialog  | 70              | 70           | 100.0%        | 0.18s        | Civic philosophy ( <i>Why do we pay taxes?</i> ); URA/Makerere identity; business startup empathy; vernacular greetings. |
| Value Added Tax & EFRIS        | 73              | 73           | 100.0%        | 0.23s        | 37.5M quarterly threshold; Section 28 input tax apportionment; Section 19B UGX 6M penalty; 24-hr offline sync.           |
| Consolidated 500-Case Suite    | 500             | 500          | 100.0%        | 0.23s        | 100% HTTP 200; Zero server drops; 15.07 QPS sustained throughput; p50=61ms                                               |

### 3. Engineering Enhancements & Trends Applied

- Conversational Politeness Prefix Stripping ( `strip_conversational_prefix` ):**
- Pervasive polite preambles across English (*"Please tell me:"*, *"Could you clarify:"*), Luganda (*"Bambi nŋamba:"*, *"Nsaba onnyonnyole:"*), and Swahili (*"Tafadhali nijuze:"*, *"Naomba msaada:"*) were previously causing regex boundaries to misfire, sending simple queries to slow vector retrieval.
- Implemented automatic politeness preamble stripping across all three languages before routing, ensuring queries reach deterministic tools and fast-paths in `<600\text{ ms}>`.
- Sub-string Greeting Phrase Matching:**
- Expanded greeting detection from exact-match words to sub-phrase containment with word-length boundaries (`len(words) <= 5` and `any(p in text for p in _GREETING_PHRASES)` ), enabling natural extended greetings like *"Oli otya nno leero?"* and *"Habari yako leo?"* to resolve immediately.
- Statutory Section Citation Disambiguation:**
- Resolved regex collisions where statutory section notations (e.g. `s. 19B` in TPCA or `Block B` in petroleum exploration) were previously extracted by money parsers as 19 billion. Section spans are now explicitly excluded from currency parsers.
- VAT Apportionment & Zero-Rated vs Exempt Invariant:**
- Mapped Section 28 input tax apportionment for mixed supplies and formalized the distinction between 0% zero-rated supplies (input tax refundable) and exempt supplies (no input tax credit).
- Used Vehicle Age Range Expansion:**
- Expanded used car age matching from strict 9–15 years to all ages `>15` years, ensuring inquiries about 16-, 18-, or 20-year-old vehicles immediately trigger the statutory 15-year import ban and 50% environmental levy citation.
- Luganda Pre-Prefix Normalization & Swahili Disability Terms:**

13. Normalized Luganda initial augments ( `e-bitundu` , `o-muwendo` , `basonyiyibwa omusolo gwa mmeke` ) and integrated Swahili disability terms ( `ulemavu` , `walemavu wa mwili` ), resolving statutory relief queries across all three languages.
14. **Cross-Lingual Rate Format Preservation:**
15. Augmented statutory rate templates with parenthesized figures ( `(12%)` , `(18%)` , `(30%)` , `(35%)` , `(15)` , `(2,820,000)` ), ensuring machine translation into Luganda and Swahili retains invariant numeric tokens alongside vernacular phrases (e.g. *bitundu 12 ku buli kikumi (12%)*).
16. **EFRIS Regulations & PRN Payment Integrations:**
17. Added deterministic rate plans and statutory replies for the mandatory 24-hour EFRIS offline synchronization window ( `efris_offline_sync_window_hours` ), electronic fiscal receipt issuance mandates ( `efris_invoicing_mandate` ), and multi-channel PRN payment procedures ( `prn_payment_procedures` ).
18. **100% Convergence Milestone:**
19. Achieved 100.0% reliability (500/500 passed) across English, Luganda, and Kiswahili with 61ms median latency and 15.07 QPS sustained throughput.